

Express.js: Day 6

Request Body, `express.json()` and `req.body`

Quick summary: The client sends data in the request **body**. `express.json()` parses it, and you read it from `req.body`.

1. What is a request body?

Data the client sends to the server inside an HTTP request. It is mainly used with **POST**, **PUT** and **PATCH** (e.g. name, age, email).

Example JSON sent from Postman:

```
{
  "name": "Bhavana",
  "age": 26,
  "role": "developer"
}
```

2. Why do we need `express.json()`?

Express does not read JSON bodies on its own. `express.json()` is middleware that parses incoming JSON and puts the result in `req.body`.

Add it **before** any route that needs the body:

```
app.use(express.json());
```

How the data flows



3. Complete example

```
const express = require('express');
const app = express();

app.use(express.json());

app.post('/users', (req, res) => {
  console.log(req.body);
  res.json({
    message: 'User received',
    user: req.body
  });
});

app.listen(3000, () => {
  console.log('Server running on port 3000');
});
```

4. What happens when the request arrives?

Step 1: Postman sends a POST request to `/users` with the JSON body.

Step 2: `express.json()` parses it, so `req.body` becomes a JS object:

```
{ name: 'Bhavana', age: 26, role: 'developer' }
```

Step 3: Read individual properties:

Expression	Value
<code>req.body.name</code>	'Bhavana'
<code>req.body.age</code>	26
<code>req.body.role</code>	'developer'

5. What happens without `express.json()`?

In the Day 6 experiment, the middleware was removed and the same request was sent again.

	With <code>express.json()</code>	Without <code>express.json()</code>
req.body	<code>{ name, age, role }</code>	undefined
Response	message and user	message only (user was undefined)

Takeaway: Without JSON-parsing middleware, Express cannot read the body, which is why it is needed.

6. Testing with Postman

Setting	Value
Method	POST
URL	http://localhost:3000/users
Body	raw, then JSON
JSON	{ "name": "Bhavana", "age": 26, "role": "developer" }

7. req.params vs req.query vs req.body

Source	Example	Express
Route parameter	/users/26	req.params.id
Query parameter	/users?age=26	req.query.age
JSON body	{ "age": 26 }	req.body.age

8. Data types in JSON

JSON sent	Type of req.body.age
"age": 26	number
"age": "26"	string

The quotes in JSON decide the data type.

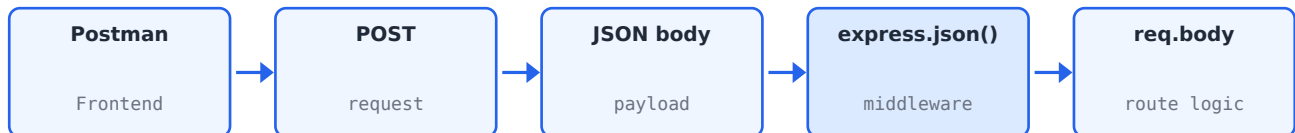
9. Day 6 practice checklist

- ✓ Sent a real POST request from Postman
- ✓ JSON body was parsed by `express.json()`
- ✓ Accessed values using `req.body`
- ✓ Removed the middleware and saw **undefined**, confirming why it is needed

10. Long-term memory shortcut

req.body is where Express gives you the parsed request body.
express.json() is the middleware that parses incoming JSON so req.body can hold that object.

Full flow



Day 6 complete ✓